

Math 220A Final

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1 Setup: Induced Representations

Let $H < G$ be a subgroup of a finite group G . In the recording of the last lecture, we have defined the functor

$$\mathrm{Ind}_H^G : \mathrm{Rep}_k G \rightarrow \mathrm{Rep}_k H \quad (1.1)$$

from (finite dimensional) representations of H to representations of G as the **right adjoint** of the restriction functor $\mathrm{Res}_H^G : \mathrm{Rep}_k G \rightarrow \mathrm{Rep}_k H$. The first part of this problem is to prove that this right adjoint exists and study some of its properties.

2 Problems:

2.1 D

Problem 1

Let $W \in \mathrm{Rep}_k(H)$ be a finite dimensional representation of H . Define a k -vector space W' as follow: Let

$$W' := \{\varphi : G \rightarrow W \mid \varphi(hg) = h \cdot \varphi(g), \quad \forall h \in H, g \in G\} \quad (2.1)$$

Next, for any $g \in G$ and $\varphi \in W'$, let

$$g \cdot \varphi : G \rightarrow W \quad (2.2)$$

denote the function sending $x \mapsto \varphi(xg)$. Show that $g \cdot \varphi \in W'$ and show that the association $(g, \varphi) \mapsto g \cdot \varphi$ defines a representation of G on W' . We shall define $\mathrm{Ind}_H^G W = W'$ as a representation of G (at the moment you should take Ind_H^G as a symbol. Later we will justify such a choice by showing this indeed provides the desired right adjoint of Res_H^G).

Solution: First, given any $\varphi \in W'$ and $g, x \in G$, one has $(g \cdot \varphi)(x) = \varphi(xg)$. Which, for all $h \in H$, one has the following:

$$(g \cdot \varphi)(hx) = \varphi(hxg) = h \cdot \varphi(xg) = h \cdot (g \cdot \varphi)(x) \quad (2.3)$$

Hence, this relation shows that $g \cdot \varphi \in W'$.

Now, to prove that the association $(g, \varphi) \mapsto g \cdot \varphi$ defines a representation, first we show it's a linear map: Given any $a, b \in k$, $\varphi, \psi \in W'$, and any $g, x \in G$, one has the following:

$$\begin{aligned}(g, a\varphi + b\psi)(x) &= g \cdot (a\varphi + b\psi)(x) = a\varphi(xg) + b\psi(xg) = a(g \cdot \varphi)(x) + b(g \cdot \psi)(x) \\ &= (a(g, \varphi) + b(g, \psi))(x)\end{aligned}\quad (2.4)$$

So, $(g, a\varphi + b\psi) = a(g, \varphi) + b(g, \psi)$, showing such association is a linear map.

Then, to show it's a group action, for any $g, g', x \in G$ and $\varphi \in W'$, one has the following:

$$(g, (g', \varphi))(x) = (g \cdot (g' \cdot \varphi))(x) = (g \cdot \varphi)(xg') = \varphi(xgg') = (gg' \cdot \varphi)(x) = (gg', \varphi)(x) \quad (2.5)$$

So, $(g, (g', \varphi)) = (gg', \varphi)$, showing such association forms a group action. Hence, this defines a group representation of G (since it's a group action, which is a group homomorphism into the automorphism group of W' ; and, it's verified that all such automorphism is a linear operator of W' , hence results in a group homomorphism $G \rightarrow \text{GL}(W')$).

2.2 D

Problem 2

Let $K < H < G$ be subgroups of G . Let W_0 be a finite-dimensional representation of K . Show that

$$\text{Ind}_K^G W_0 \cong \text{Ind}_H^G \text{Ind}_K^H W_0 \quad (2.6)$$

as G -representations.

Solution:

I. The Linear Map $T : \text{Ind}_K^G W_0 \rightarrow \text{Ind}_H^G (\text{Ind}_K^H W_0)$:

First, notice that for all $\varphi : G \rightarrow W_0$ in $\text{Ind}_K^G W_0$, its restriction onto H has $\varphi|_H \in \text{Ind}_K^H W_0$, since for all $y \in H$ and $k \in K < H$, one has $ky \in H$, hence $\varphi|_H(ky) = \varphi(ky) = k \cdot \varphi(y) = k \cdot \left(\varphi|_H(y) \right)$.

Which, based on this property, one can define a map $T_\varphi : G \rightarrow \text{Ind}_K^H W_0$ by $T_\varphi(g) = (g \cdot \varphi)|_H$. Notice that $T_\varphi \in \text{Ind}_H^G (\text{Ind}_K^H W_0)$: For any $h, y \in H$ and $g \in G$, one has the following result:

$$T_\varphi(hg)(y) = ((hg) \cdot \varphi)|_H(y) = ((hg) \cdot \varphi)(y) = (h \cdot (g \cdot \varphi))(y) = (g \cdot \varphi)(yh) \quad (2.7)$$

Which, since $y, h \in H$, then $yh \in H$ also. So, one results in the following:

$$T_\varphi(hg)(y) = (g \cdot \varphi)(yh) = (g \cdot \varphi)|_H(yh) = h \cdot \left((g \cdot \varphi)|_H(y) \right) = h \cdot T_\varphi(g)(y) \quad (2.8)$$

Hence, one deduces that $T_\varphi(hg) = h \cdot T_\varphi(g)$, which satisfies the property of $\text{Ind}_H^G (\text{Ind}_K^H W_0)$.

Finally, based on this property, define the map $T : \text{Ind}_K^G W_0 \rightarrow \text{Ind}_H^G (\text{Ind}_K^H W_0)$ by $T(\varphi) := T_\varphi$. Which, for all $a, b \in k$, and $\varphi, \psi \in \text{Ind}_K^G W_0$, it satisfies the following for all $g \in G$:

$$\begin{aligned}T(a\varphi + b\psi)(g) &= T_{a\varphi + b\psi}(g) = (g \cdot (a\varphi + b\psi))|_H = (a(g \cdot \varphi) + b(g \cdot \psi))|_H \\ &= a \left((g \cdot \varphi)|_H \right) + b \left((g \cdot \psi)|_H \right) = aT_\varphi(g) + bT_\psi(g) = (aT(\varphi) + bT(\psi))(g)\end{aligned}\quad (2.9)$$

Hence, we conclude that $T(a\varphi + b\psi) = aT(\varphi) + bT(\psi)$ as maps in $\text{Ind}_H^G(\text{Ind}_K^H W_0)$, showing T is indeed a linear map.

II. T is a Linear Isomorphism:

For this, we'll define an inverse. Given any map $\Phi \in \text{Ind}_H^G(\text{Ind}_K^H W_0)$, it is an association $\Phi(g) = \varphi_g \in \text{Ind}_K^H W_0$, such that for all $h \in H$, one has $\Phi(hg) = \varphi_{hg} = h \cdot (\varphi_g) = h \cdot \Phi(g)$ (by the definition of Ind_H^G). Which, for observation purpose, suppose there exists $\varphi \in \text{Ind}_K^G W_0$, such that $T(\varphi) = T_\varphi$ matches such association Φ (i.e. for all $g \in G$, one has $T_\varphi(g) = (g \cdot \varphi)|_H = \varphi_g = \Phi(g)$), then φ must satisfy $\varphi(g) = \varphi(1g) = (g \cdot \varphi)(1) = \varphi_g(1) = \Phi(g)(1)$ (here $1 \in G$ is the identity, also in H).

So, define the map $\varphi : G \rightarrow W_0$, by $\varphi(g) := \varphi_g(1)$. We claim that $\varphi \in \text{Ind}_K^G W_0$: Given any $g \in G$ and $k \in K$, one has the following (Note: Since $k \in K < H$, so the above property of φ_g can be used):

$$\varphi(kg) = \varphi_{kg}(1) = (k \cdot \varphi_g)(1) = \varphi_g(k) = k \cdot (\varphi_g(1)) = k \cdot \varphi(g) \quad (2.10)$$

(Note the third equality is based on the fact that $\varphi_g \in \text{Ind}_K^H W_0$). Which, φ satisfies the property of elements in $\text{Ind}_K^G W_0$. Also, notice that T_φ satisfies the following for all $g \in G$ and $h \in H$:

$$(T_\varphi(g))(h) = (g \cdot \varphi)|_H(h) = (g \cdot \varphi)(h) = \varphi(hg) = \varphi_{hg}(1) = (h \cdot \varphi_g)(1) = \varphi_g(h) \quad (2.11)$$

This shows that $T_\varphi(g) = \varphi_g = \Phi(g)$ for all $g \in G$, achieving the fact that $T_\varphi = \Phi$.

Now, we define the map $S : \text{Ind}_H^G(\text{Ind}_K^H W_0) \rightarrow \text{Ind}_K^G W_0$, by $S(\Phi) := \varphi$ that is defined above, and claim that $S = T^{-1}$: Given any $\varphi \in \text{Ind}_K^G W_0$, one has the following:

$$S \circ T(\varphi) = S(T_\varphi), \quad \forall g \in G, \quad S(T_\varphi)(g) = T_\varphi(g)(1) = (g \cdot \varphi)|_H(1) = (g \cdot \varphi)(1) = \varphi(g) \quad (2.12)$$

This shows that $S \circ T(\varphi) = \varphi \in \text{Ind}_K^G W_0$. Also, Based on the result proven before, any $\Phi \in \text{Ind}_H^G(\text{Ind}_K^H W_0)$ has $\varphi = S(\Phi)$ satisfies $T(\varphi) = T_\varphi = \Phi$. Hence, these two equality shows that $S \circ T = \text{id}_{\text{Ind}_K^G W_0}$, while $T \circ S = \text{id}_{\text{Ind}_H^G(\text{Ind}_K^H W_0)}$, showing S is the inverse of T . Hence, $\text{Ind}_K^G W_0 \cong \text{Ind}_H^G(\text{Ind}_K^H W_0)$ as k -vector spaces.

III. T is a G -representation Isomorphism:

It suffices to show that T is a G -representation morphism. Given any $g, g' \in G$ and any $\varphi \in \text{Ind}_K^G W_0$, one has the following action on $h \in H$:

$$\begin{aligned} (T(g \cdot \varphi))(g') &= T_{g \cdot \varphi}(g') = (g' \cdot (g \cdot \varphi))|_H = ((g'g) \cdot \varphi)|_H \\ (g \cdot T(\varphi))(g') &= (g \cdot T_\varphi)(g') = T_\varphi(g'g) = ((g'g) \cdot \varphi)|_H \end{aligned} \quad (2.13)$$

This shows that $T(g \cdot \varphi) = g \cdot T(\varphi)$ as map in $\text{Ind}_H^G(\text{Ind}_K^H W_0)$. Hence, T is a G -representation morphism, hence a G -representation isomorphism.

As a consequence, $\text{Ind}_K^G W_0 \cong \text{Ind}_H^G(\text{Ind}_K^H W_0)$ as G -representations.

2.3 D

Problem 3

Choose representatives $\{x_1, \dots, x_l\}$ of G/H , and write $G = \bigsqcup_{i=1}^l Hx_i$ (we may and will always assume that $x_1 = 1$). Let $W^{(i)} \subset \text{Ind}_H^G W$ denote the subspace

$$W^{(i)} := \{\varphi \in \text{Ind}_H^G W \mid \varphi(x) = 0 \text{ if } x \notin Hx_i\} \quad (2.14)$$

Show that each $W^{(i)}$ has dimension equal to $\dim_k W$. Show that as vector spaces, we have

$$\text{Ind}_H^G W \cong \bigoplus_{i=1}^l W^{(i)} \quad (2.15)$$

Conclude that $\text{Ind}_H^G W$ has dimension $[G : H] \cdot \dim_k W$ as a vector space over k .

Solution:**I. Dimension of $W^{(i)}$:**

First fix an arbitrary representative x_j . Let $n = \dim_k W$, choose a basis $w_1, \dots, w_n \in W$, and for each $i \in \{1, \dots, n\}$, define $\varphi_i : G \rightarrow W$ as follow:

$$\varphi_i(x) = \begin{cases} h \cdot w_i, & x = hx_j \in Hx_j \\ 0, & \text{otherwise} \end{cases} \quad (2.16)$$

Notice that given any $g \in G$ and $h \in H$, there are two cases:

- If $g \in Hx_j$ (say $g = h'x_j$), then $hg \in Hx_j$, so $\varphi_i(hg) = (hg) \cdot w_i = h \cdot (h' \cdot w_i) = h \cdot (\varphi_i(g))$.
- Else if $g \notin Hx_j$, then $hg \notin Hx_j$, showing $\varphi_i(hg) = 0 = h \cdot 0 = h \cdot (\varphi_i(g))$.

Hence, regardless of the case one has $\varphi_i(hg) = h \cdot \varphi_i(g)$ for all $g \in G$ and $h \in H$, showing $\varphi_i \in \text{Ind}_H^G W$ (in particular in $W^{(i)}$).

Now, we claim that $\varphi_1, \dots, \varphi_n$ forms a basis for $W^{(j)}$:

- For spanning property, given any $\varphi \in W^{(j)}$, let $\varphi(x_j) = \sum_{i=1}^n a_i w_i$, which we claim that $\varphi = \sum_{i=1}^n a_i \varphi_i$. For all $g \in G$, if $g \notin Hx_j$, it's clear that $\varphi(g) = 0 = \sum_{i=1}^n a_i \varphi_i(g)$; else if $g \in Hx_j$, there exists $h \in H$ such that $g = hx_j$. Hence, it satisfies the following:

$$\varphi(g) = \varphi(hx_j) = h \cdot \varphi(x_j) = \sum_{i=1}^n a_i (h \cdot w_i) = \sum_{i=1}^n a_i \varphi_i(g) \quad (2.17)$$

So, one concludes that $\varphi = \sum_{i=1}^n a_i \varphi_i$, showing $\{\varphi_1, \dots, \varphi_n\}$ form a spanning set.

- For linear independence, suppose $\sum_{i=1}^n a_i \varphi_i = 0$, plugin x_j , one has $\sum_{i=1}^n a_i \varphi_i(x_j) = \sum_{i=1}^n a_i w_i = 0$, so by linear independence of $\{w_1, \dots, w_n\}$, it enforces all $a_i = 0$. Hence, the list $\{\varphi_1, \dots, \varphi_n\}$ is linearly independent.

This shows that $\{\varphi_1, \dots, \varphi_n\}$ indeed forms a basis for $W^{(j)}$, hence $\dim_k W^{(j)} = n = \dim_k W$.

II. Direct Sum Decomposition of $\text{Ind}_H^G W$:

For all $\varphi \in \text{Ind}_H^G W$, let each $j \in \{1, \dots, l\}$ corresponds to a map $\varphi^{(j)} : G \rightarrow W$ as follow:

$$\varphi^{(j)}(x) = \begin{cases} \varphi(x), & x \in Hx_j \\ 0, & \text{otherwise} \end{cases} \quad (2.18)$$

Which, for all $g \in G$ and $h \in H$, if $g \in Hx_j$, one has $hg \in Hx_j$, hence $\varphi^{(j)}(hg) = \varphi(hg) = h \cdot \varphi(g) = h \cdot \varphi^{(j)}(g)$; else if $g \notin Hx_j$, then $hg \notin Hx_j$, showing $\varphi^{(j)}(hg) = 0 = h \cdot 0 = h \cdot \varphi^{(j)}(g)$. So, one has $\varphi^{(j)} \in \text{Ind}_H^G W$, in particular $\varphi^{(j)} \in W^{(j)}$ by definition.

Notice that $\varphi = \sum_{j=1}^l \varphi^{(j)}$, since for all $g \in G$, there exists a unique $i \in \{1, \dots, l\}$, such that $g \in Hx_i$. So, one has $\varphi^{(j)}(g) = \delta_{ij} \varphi(g)$ (since for $i \neq j$ it's 0, for $i = j$ it's $\varphi(g)$). Hence, $\varphi(g) = \sum_{j=1}^l \delta_{ij} \varphi(g) = \sum_{j=1}^l \varphi^{(j)}(g)$, showing $\varphi = \sum_{j=1}^l \varphi^{(j)}$ (since $g \in G$ is arbitrary). This proves that $\text{Ind}_H^G W = \sum_{j=1}^l W^{(j)}$ (since each $\varphi \in \text{Ind}_H^G W$ can be expressed as sums of elements in each $W^{(j)}$).

Now, to show it's indeed a direct sum, suppose for each $j \in \{1, \dots, l\}$, the corresponding $\varphi^{(j)} \in W^{(j)}$ satisfies $\sum_{j=1}^l \varphi^{(j)} = 0$, then fix any representative x_i , any $g \in Hx_i$ has $\varphi^{(j)}(g) = 0$ (if $i \neq j$), so one has the following:

$$\sum_{j=1}^l \varphi^{(j)}(g) = \varphi^{(i)}(g) = 0 \quad (2.19)$$

Then, since $g \in Hx_i$ is arbitrary, this shows that $\varphi^{(i)} = 0$ (since the only possible nonzero values are from Hx_i , but now they're all zero). So, each component must be zero, showing that the $W^{(j)}$ does form a direct sum. Hence, $\text{Ind}_H^G W = \bigoplus_{i=1}^l W^{(i)}$.

As a result, since each $\dim_k W^{(i)} = \dim_k W$, and the number of representatives $l = [G : H]$, one has $\dim_k(\text{Ind}_H^G W) = \sum_{i=1}^l \dim_k W^{(i)} = l \cdot \dim_k W = [G : H] \cdot \dim_k W$.

2.4 D

Problem 4

Next, for any $w \in W$ and $1 \leq i \leq l$, let us consider the following function

$$\varphi_w^{(i)} : G \rightarrow W \quad (2.20)$$

which sends an element of the form hx_i (where $h \in H$) to $h \cdot w$ and all other elements (namely ones of the form hx_j for $j \neq i$) in G to 0. Show that $\varphi_w^{(i)} \in W^{(i)}$ and in fact we have

$$W^{(i)} = \left\{ \varphi_w^{(i)} \mid w \in W \right\} \quad (2.21)$$

as subspaces of $\text{Ind}_H^G W$.

Solution: First, to show that $\varphi_w^{(i)} \in W^{(i)}$, it suffices to show that $\varphi_w^{(i)} \in \text{Ind}_H^G W$ (since by description it already satisfies the requirement that $\varphi(g) = 0$ for $g \notin Hx_i$, so it just needs to be in the space). Given any $g \in G$ and $h \in H$, if $g \in Hx_i$, there exists $h' \in H$ where $g = h'x_i$, then one has $\varphi_w^{(i)}(hg) = \varphi^{(i)}((hh')x_i) = (hh') \cdot w = h \cdot (h' \cdot w) = h \cdot \varphi_w^{(i)}(g)$, showing that $\varphi_w^{(i)}$ satisfies the property for being in the space, hence $\varphi_w^{(i)} \in \text{Ind}_H^G W$, which is also in $W^{(i)}$.

Now, for every $\varphi \in W^{(i)}$, consider the vector $w = \varphi(x_i)$: For any $g \in Hx_i$, there exists $h \in H$ such that $g = hx_i$, hence one has the following:

$$\varphi(g) = \varphi(hx_i) = h \cdot \varphi(x_i) = h \cdot w = \varphi_w^{(i)}(x_i) \quad (2.22)$$

Else if $g \notin Hx_i$, it's clear that $\varphi(g) = 0 = \varphi_w^{(i)}(g)$. So, one has $\varphi = \varphi_w^{(i)}$, where $w = \varphi(x_i)$.

This shows that $W^{(i)} = \left\{ \varphi_w^{(i)} \mid w \in W \right\}$ for each index i .

2.5 D

Problem 5

Show that for all $\varphi \in \text{Ind}_H^G W$, we have

$$\varphi = \sum_{i=1}^l \varphi_{\varphi(x_i)}^{(i)} \quad (2.23)$$

Solution: For all $g \in G$, there exists a unique index $i \in \{1, \dots, l\}$, such that $g \in Hx_i$, hence one has a unique $h \in H$ also such that $g = hx_i$. Which, for all index $i \neq j$, one has $\varphi_{\varphi(x_i)}^{(i)}(g) = 0$ (since $g \notin Hx_i$), and for $i = j$, one has the following:

$$\varphi_{\varphi(x_j)}^{(j)}(g) = \varphi_{\varphi(x_j)}^{(j)}(hx_j) = h \cdot \varphi(x_j) = \varphi(hx_j) = \varphi(g) \quad (2.24)$$

So, one concludes that $\varphi(g) = \varphi_{\varphi(x_j)}^{(j)}(g) = \sum_{i=1}^l \varphi_{\varphi(x_i)}^{(i)}(g)$. Hence, with $g \in G$ being arbitrary, $\varphi = \sum_{i=1}^l \varphi_{\varphi(x_i)}^{(i)}$ being the unique decomposition for the previous direct sums.

2.6 D

Problem 6

Now let $W \in \text{Rep}_k H$ and let $V \in \text{Rep}_k G$. Consider the (evaluation) map

$$e : \text{Ind}_H^G W \rightarrow W \quad (2.25)$$

sending $\varphi \mapsto \varphi(1)$ as a map between vector spaces. Show that e is in fact a map between H -representations.

Solution: First, let's verify that e is a linear map: For any $a, b \in k$ and $\varphi, \psi \in \text{Ind}_H^G W$, one has the following:

$$e(a\varphi + b\psi) = (a\varphi + b\psi)(1) = a\varphi(1) + b\psi(1) = a \cdot e(\varphi) + b \cdot e(\psi) \quad (2.26)$$

So, it's a linear map. Now, for all $h \in H$ and $\varphi \in \text{Ind}_H^G W$, one has the following:

$$e(h \cdot \varphi) = (h \cdot \varphi)(1) = \varphi(1h) = h \cdot (\varphi(1)) = h \cdot (e(\varphi)) \quad (2.27)$$

This shows that e is an H -representation morphism.

2.7 D

Problem 7

Use Part (6) to show that Ind_H^G is a right adjoint of Res_H^G .

Solution: Given any $V \in \text{Rep}_k(G)$ and $W \in \text{Rep}_k(H)$, notice that the evaluation map $e : \text{Ind}_H^G W \rightarrow W$ in **Problem 6**. Then, for any G -representation morphism $T \in \text{Hom}_G(V, \text{Ind}_H^G W)$ (which is also

an H -representation morphism by restricting to only the H -action), then $e \circ T : V \rightarrow W$ is in fact a H -representation morphism (here denote $\text{Res}_H^G V := V$ as vector space, with only the H -representation structure). Hence, the map $e \circ _ : \text{Hom}_G(V, \text{Ind}_H^G W) \rightarrow \text{Hom}_H(\text{Res}_H^G V, W)$ by $T \mapsto e \circ T$ is a well-defined map (in fact a k -linear map).

Now, the goal is to show that $e \circ _$ is a bijection by constructing an inverse: Given any H -representation morphism $S \in \text{Hom}_H(\text{Res}_H^G V, W)$, for all $v \in V$, define a map $\varphi_v : G \rightarrow W$, by $\varphi_v(x) := S(x \cdot v)$. Notice that for any $x \in G$, $h \in H$, this map satisfies the following:

$$\varphi_v(hx) = S((hx) \cdot v) = S(h \cdot (x \cdot v)) = h \cdot S(x \cdot v) = h \cdot \varphi_v(x) \quad (2.28)$$

Hence, this implies $\varphi_v \in \text{Ind}_H^G W$. Which, define a map $I_S : V \rightarrow \text{Ind}_H^G W$ by $I_S(v) := \varphi_v$, we can show it's in fact a G -representation morphism:

- First, given any $a, b \in k$ and $u, v \in V$, the map satisfies the following:

$$\begin{aligned} \forall x \in G, I_S(au + bv)(x) &= \varphi_{au+bv}(x) = S(x \cdot (au + bv)) = aS(x \cdot u) + bS(x \cdot v) \\ &= a\varphi_u(x) + b\varphi_v(x) = (aI_S(u) + bI_S(v))(x) \end{aligned} \quad (2.29)$$

Hence, $I_S(au + bv) = aI_S(u) + bI_S(v)$, showing it's a linear map.

- Given any $g, x \in G$ and $v \in V$, one has the map satisfies the following:

$$\begin{aligned} I_S(g \cdot v)(x) &= \varphi_{g \cdot v}(x) = S(x \cdot (g \cdot v)) = S((xg) \cdot v) \\ &= \varphi_v(xg) = (g \cdot \varphi_v)(x) = (g \cdot I_S(v))(x) \end{aligned} \quad (2.30)$$

This shows that $I_S(g \cdot v) = g \cdot I_S(v)$, showing I_S is a G -representation morphism,

With these properties, define a map $f : \text{Hom}_H(\text{Res}_H^G V, W) \rightarrow \text{Hom}_G(V, \text{Ind}_H^G W)$ by $f(S) := I_S$ defined above. Which, we claim that $e \circ _$ and f are mutual inverse.

- Given any $T \in \text{Hom}_G(V, \text{Ind}_H^G W)$, any $v \in V$ has $T(v) \in \text{Ind}_H^G W$, which for all $x \in G$, one has the following:

$$T(v)(x) = T(v)(1x) = (x \cdot T(v))(1) = e(x \cdot T(v)) \quad (2.31)$$

Also, $f(e \circ T) = I_{e \circ T}$ has all $v \in V$ satisfies the following for all $x \in G$:

$$I_{e \circ T}(v)(x) = e \circ T(x \cdot v) = e(x \cdot T(v)) \quad (2.32)$$

Hence, one gets that $T(v) = I_{e \circ T}(v)$, showing $T = I_{e \circ T} = f(e \circ T)$. Hence, $f \circ (e \circ _)$ is identity on $\text{Hom}_G(V, \text{Ind}_H^G W)$.

- Given any $S \in \text{Hom}_H(\text{Res}_H^G V, W)$, one has $f(S) = I_S \in \text{Hom}_G(V, \text{Ind}_H^G W)$. Which, for any $v \in V$, $e \circ I_S$ satisfies the following:

$$e \circ I_S(v) = e(\varphi_v) = \varphi_v(1) = S(1 \cdot v) = S(v) \quad (2.33)$$

Hence, this shows that $e \circ (f(S)) = e \circ I_S = S$, showing that $(e \circ _) \circ f$ is identity on $\text{Hom}_H(\text{Res}_H^G V, W)$.

Hence, these two properties show that $e \circ _ : \text{Hom}_G(V, \text{Ind}_H^G W) \rightarrow \text{Hom}_H(\text{Res}_H^G V, W)$ and $f : \text{Hom}_H(\text{Res}_H^G V, W) \rightarrow \text{Hom}_G(V, \text{Ind}_H^G W)$ are mutual inverse, hence Ind_H^G is a right adjoint of Res_H^G .

2.8 ND

Problem 8

Consider the following special example: Let V_{st} denote the irreducible 2-dimensional representation of S_3 . Compute the character of $\text{Ind}_{S_3}^{S_4} V_{st}$, compute the character table of S_4 , then decompose $\text{Ind}_{S_3}^{S_4} V_{st}$ as direct sums of irreducible representations of S_4 .

2.9 HD (calculate example for S3 induce S4)

Problem 9

For any function $f : H \rightarrow \mathbb{C}$, let us define a function $\tilde{f} : G \rightarrow \mathbb{C}$ by

$$\tilde{f}(x) = \begin{cases} f(x) & \text{if } x \in H \\ 0 & \text{otherwise} \end{cases} \quad (2.34)$$

(Without using Part (7)). Show that the character of $\text{Ind}_H^G W$ is given by

$$\chi_{\text{Ind}_H^G W}(x) = \frac{1}{\#H} \sum_{g \in G} \tilde{\chi}_W(gxg^{-1}) \quad (2.35)$$

Compute the character of $\text{Ind}_{S_3}^{S_4} V_{\text{st}}$ one more time using this formula.

Solution:

I. Choose a Basis for $\text{Ind}_H^G W$:

First, let's consider a basis: in **Problem 5** it has been verified that all $\varphi \in \text{Ind}_H^G W$ has decomposition $\varphi = \sum_{i=1}^l \varphi_{\varphi(x_i)}^{(i)}$, each $\varphi_{\varphi(x_i)}^{(i)} \in W^{(i)}$ has $\varphi_{\varphi(x_i)}^{(i)}(hx_i) = h \cdot \varphi(x_i)$ for all $h \in H$ (this decomposition is based on $W = \bigoplus_{i=1}^l W^{(i)}$, each $W^{(i)}$ associates to the coset Hx_i).

On the other hand, for each $a, b \in k$ and $u, w \in W$, the element $\varphi_{au+bw}^{(i)}(hx_i) = h \cdot (au + bw) = a(h \cdot u) + b(h \cdot w) = a\varphi_u^{(i)}(hx_i) + b\varphi_w^{(i)}(hx_i)$ for all $h \in H$ (with $\varphi_{au+bw}^{(i)}(hx_j) = 0 = a\varphi_u^{(i)}(hx_j) + b\varphi_w^{(i)}(hx_j)$ for $j \neq i$), showing that $\varphi_{au+bw}^{(i)} = a\varphi_u^{(i)} + b\varphi_w^{(i)}$.

So, pick a basis $w, \dots, w_n \in W$, then $\{\varphi_{w_j}^{(i)}\}_{\substack{1 \leq i \leq l \\ 1 \leq j \leq n}}$ is a basis for $\text{Ind}_H^G W$, since each $\varphi \in W$ has $\varphi(x_i) = \sum_{j=1}^n a_{ij}w_j$ for unique collections of $a_{ij} \in k$, hence one has $\varphi = \sum_{i=1}^l \varphi_{\varphi(x_i)}^{(i)} = \sum_{i=1}^l \sum_{j=1}^n a_{ij}\varphi_{w_j}^{(i)}$, with each coefficient a_{ij} being uniquely determined by each $\varphi(x_i)$.

II. G -Action on the Basis:

Given any $x \in G$, it induces a permutation on the right cosets $\{Hx_i\}_{1 \leq i \leq l}$, say $\sigma \in S_l$ has $H(x_i x) = Hx_{\sigma(i)}$. In particular each $x_i x = h_i x_{\sigma(i)}$ for some fixed $h_i \in H$. Then, for each basis element $\varphi_{w_j}^{(i)}$, the action of x is as follow:

$$\forall x_r \in G, \quad (x \cdot \varphi_{w_j}^{(i)})(x_r) = \varphi_{w_j}^{(i)}(x_r x) = \varphi_{w_j}^{(i)}(h_r x_{\sigma(i)}) = \delta_{i, \sigma(i)}(h_r \cdot w_j) \quad (2.36)$$

So, this indicates that $x \cdot \varphi_{w_j}^{(i)} = \sum_{r=1}^l \varphi_{\delta_{i, \sigma(i)}(h_r \cdot w_j)}^{(r)}$, hence the $\varphi_{w_j}^{(i)}$ -component in this expression is $\delta_{i, \sigma(i)} a_{jj}^{(i)}$, with each h_i (as a linear operator on W) has matrix $\left(a_{jk}^{(i)}\right)_{1 \leq j, k \leq n}$ (since within $\varphi_{\delta_{i, \sigma(i)}(h_i \cdot w_j)}^{(i)}$, the $\varphi_{w_j}^{(i)}$ -component is the w_j -component of $\delta_{i, \sigma(i)}(h_i \cdot w_j)$, given by $\delta_{i, \sigma(i)} a_{jj}^{(i)}$). As a result, one has the following:

$$\chi_{\text{Ind}_H^G W}(x) = \sum_{i=1}^l \sum_{j=1}^n \delta_{i, \sigma(i)} a_{jj}^{(i)} = \sum_{i=1}^l \delta_{i, \sigma(i)} \text{tr} \left(a_{jk}^{(i)} \right) = \sum_{i=1}^l \delta_{i, \sigma(i)} \chi_W(h_i) \quad (2.37)$$

III. Rewrite of the Character:

Finally, consider the term $\sum_{g \in G} \tilde{\chi}_W(gxg^{-1})$: Since $G = \bigsqcup_{i=1}^l Hx_i$, then $\sum_{g \in G} \tilde{\chi}_W(gxg^{-1}) = \sum_{i=1}^l \sum_{h \in H} \tilde{\chi}((hx_i)x(x_i^{-1}h^{-1})) = \sum_{i=1}^l \sum_{h \in H} \tilde{\chi}_W(hh_i x_{\sigma(i)} x_i^{-1} h^{-1})$ (Note: by our definition $x_i x = h_i x_{\sigma(i)}$). Which, $hh_i x_{\sigma(i)} x_i^{-1} h^{-1} \in H$ iff $x_{\sigma(i)} x_i^{-1} \in H$ iff $Hx_{\sigma(i)} = Hx_i$ as coset (which by the

property of each x_i being distinct representative, $Hx_{\sigma(i)} = Hx_i$ enforces $\sigma(i) = i$, hence one gets the following:

$$\tilde{\chi}_W(hh_i x_{\sigma(i)} x_i^{-1} h^{-1}) = \begin{cases} \chi_W(hh_i x_i x_i^{-1} h^{-1}) = \chi_W(hh_i h^{-1}) = \chi_W(h_i) & \sigma(i) = i \\ 0 & \text{otherwise} \end{cases} \quad (2.38)$$

Hence, we end up with the following:

$$\sum_{g \in G} \tilde{\chi}_W(gxg^{-1}) = \sum_{i=1}^l \sum_{h \in H} \tilde{\chi}_W(hh_i x_{\sigma(i)} x_i^{-1} h^{-1}) = (\#H) \sum_{i=1}^l \delta_{i, \sigma(i)} \chi_W(h_i) \quad (2.39)$$

Substituting into the term we got from **II**, we get the desired formula:

$$\chi_{\text{Ind}_H^G W}(x) = \sum_{i=1}^l \delta_{i, \sigma(i)} \chi_W(h_i) = \frac{1}{\#H} \sum_{g \in G} \tilde{\chi}_W(gxg^{-1}) \quad (2.40)$$

IV. Special Case $\text{Ind}_{S_3}^{S_4} V_{\text{st}}$:

2.10 D

Problem 10

Let $\chi : H \rightarrow \mathbb{C}$ be any class function on H , **define** a new class function $\text{Ind } \chi : G \rightarrow \mathbb{C}$ on G by

$$\text{Ind}_\chi(x) := \frac{1}{\#H} \sum_{g \in G} \tilde{\chi}(g x g^{-1}) \quad (2.41)$$

Show that for any class function χ on H and ψ on G , we have

$$\langle \text{Res } \psi, \chi \rangle_H = \langle \psi, \text{Ind}_\chi \rangle_G \quad (2.42)$$

Solution: First, notice that the left side satisfies the following:

$$\langle \text{Res } \psi, \chi \rangle_H = \frac{1}{\#H} \sum_{h \in H} \text{Res } \psi(h) \overline{\chi(h)} = \frac{1}{\#H} \sum_{h \in H} \psi(h) \overline{\chi(h)} \quad (2.43)$$

Then, based on the definition of Ind_χ , it satisfies the following:

$$\langle \psi, \text{Ind}_\chi \rangle_G = \frac{1}{\#G} \sum_{x \in G} \psi(x) \overline{\text{Ind}_\chi(x)} = \frac{1}{\#G} \sum_{x \in G} \cdot \frac{1}{\#H} \sum_{g \in G} \psi(x) \overline{\tilde{\chi}(g x g^{-1})} \quad (2.44)$$

Notice that since ψ is a class function, any $x, g \in G$ satisfies $\psi(x) = \psi(g x g^{-1})$. Hence, one has the following:

$$\langle \psi, \text{Ind}_\chi \rangle_G = \frac{1}{\#G} \sum_{g \in G} \frac{1}{\#H} \sum_{x \in G} \psi(g x g^{-1}) \overline{\tilde{\chi}(g x g^{-1})} \quad (2.45)$$

Since fixing $g \in G$, when running through $x \in G$, the collection $g x g^{-1}$ includes all elements of G (exactly once), and if $g x g^{-1} \notin H$, one has $\tilde{\chi}(g x g^{-1}) = 0$, hence the above reduces to:

$$\begin{aligned} \langle \psi, \text{Ind}_\chi \rangle_G &= \frac{1}{\#G} \sum_{g \in G} \frac{1}{\#H} \sum_{\substack{x \in G \\ g x g^{-1} \in H}} \psi(g x g^{-1}) \overline{\tilde{\chi}(g x g^{-1})} = \frac{1}{\#G} \sum_{g \in G} \frac{1}{\#H} \sum_{h \in H} \psi(h) \overline{\chi(h)} \\ &= \frac{1}{\#G} \sum_{g \in G} \langle \text{Res } \psi, \chi \rangle_H = \langle \text{Res } \psi, \chi \rangle_H \end{aligned} \quad (2.46)$$

So, this concludes that $\langle \psi, \text{Ind}_\chi \rangle_G = \langle \text{Res } \psi, \chi \rangle_H$.

2.11 D

Problem 11

Use the previous parts to give another proof of the adjunction in (7).

Solution: Here we'll discuss the case over $k = \mathbb{C}$ for the proper use of the inner product in (10).

First, based on the definition in **Problem 10**, notice that give χ_W as the character of W (as H -representations), one has the following:

$$\text{Ind}_{\chi_W}(x) = \frac{1}{\#} \sum_{g \in G} \tilde{\chi}_W(gxg^{-1}) = \chi_{\text{Ind}_H^G W}(x) \quad (2.47)$$

Hence, for any $V \in \text{Rep}_{\mathbb{C}}(G)$ and $W \in \text{Rep}_{\mathbb{C}}(H)$, the formula in **Problem 10** provides $\langle \chi_V, \chi_{\text{Ind}_H^G W} \rangle_G = \langle \chi_V, \chi_{\text{Ind}_H^G W}, \text{Ind}_{\chi_W} \rangle_G = \langle \text{Res}_H^G \chi_V, \chi_W \rangle_H = \langle \chi_{\text{Res}_H^G V}, \chi_W \rangle_H$.

Now, let V_i denotes the (finite dimensional) irreducible G -representations over $\mathbb{C}$, and W_j denotes the (finite dimensional) irreducible H -representations over $\mathbb{C}$. Then, given $V \cong \bigoplus_i V_i^{k_i}$ and $W \cong \bigoplus_j W_j^{l_j}$ as representations of G and H respectively, one has the following:

$$\text{Hom}_G(V, \text{Ind}_H^G W) \cong \text{Hom}_G\left(\bigoplus_i V_i^{k_i}, \bigoplus_j \text{Ind}_H^G W_j^{l_j}\right) \cong \bigoplus_i \left(\bigoplus_j \text{Hom}_G(V_i, \text{Ind}_H^G W_j)^{k_i l_j}\right) \quad (2.48)$$

$$\text{Hom}_H(\text{Res}_H^G V, W) \cong \text{Hom}_H\left(\bigoplus_i \text{Res}_H^G V_i, \bigoplus_j W_j^{l_j}\right) \cong \bigoplus_i \left(\bigoplus_j \text{Hom}_H(\text{Res}_H^G V_i, W_j)^{k_i l_j}\right) \quad (2.49)$$

Here, it uses the following two observations:

- $\text{Res}_H^G(V_i \oplus V_j) \cong \text{Res}_H^G V_i \oplus \text{Res}_H^G V_j$, since $g \cdot (v_i, v_j) = (g \cdot v_i, g \cdot v_j)$ for all $g \in G$, $v_i \in V_i$ and $v_j \in V_j$, by definition.
- $\text{Ind}_H^G(W_i \oplus W_j) = \{\varphi : G \rightarrow W_i \oplus W_j \mid \varphi(hg) = h \cdot \varphi(g), \forall h \in H, g \in G\}$. Which, let $\varphi_i(g) \in W_i$, $\varphi_j(g) \in W_j$ be the unique decomposition of $\varphi(g)$, say $\varphi(g) = \varphi_i(g) + \varphi_j(g)$, then one has $\varphi_i(hg) + \varphi_j(hg) = \varphi(hg) = h \cdot \varphi(g) = h \cdot \varphi_i(g) + h \cdot \varphi_j(g)$, with $\varphi_i(hg), h \cdot \varphi_i(g) \in W_i$ and $\varphi_j(hg), h \cdot \varphi_j(g) \in W_j$.

This demonstrates that $\varphi_i(hg) = h \cdot \varphi_i(g)$ and $\varphi_j(hg) = h \cdot \varphi_j(g)$ (by the property of direct sum), showing $\varphi_i \in \text{Ind}_H^G W_i$ and $\varphi_j \in \text{Ind}_H^G W_j$. Which, $\varphi = \varphi_i + \varphi_j$ for unique $\varphi_i \in \text{Ind}_H^G W_i$ and $\varphi_j \in \text{Ind}_H^G W_j$ (since by direct sum each $\varphi(g) = \varphi_i(g) + \varphi_j(g)$ must be unique), and each $\varphi \in \text{Ind}_H^G W_i$ and $\psi \in \text{Ind}_H^G W_j$ has $\varphi + \psi \in \text{Ind}_H^G(W_i \oplus W_j)$ by definition (since $\text{im}(\varphi) \subset W_i \hookrightarrow W_i \oplus W_j$, and $\text{im}(\psi) \subset W_j \hookrightarrow W_i \oplus W_j$). Which, this concludes that $\text{Ind}_H^G(W_i \oplus W_j) \cong \text{Ind}_H^G W_i \oplus \text{Ind}_H^G W_j$.

Then, using induction one can show that $\text{Res}_H^G \bigoplus_i V_i^{k_i} \cong \bigoplus_i (\text{Res}_H^G V_i)^{k_i}$, and $\text{Ind}_H^G \bigoplus_j W_j^{l_j} \cong \bigoplus_j (\text{Ind}_H^G W_j)^{l_j}$.

Hence, it suffices to show $\text{Hom}_G(V_i, \text{Res}_H^G W_j) \cong \text{Hom}_H(\text{Res}_H^G V_i, W_j)$.

Let $\text{Res}_H^G V_i \cong \bigoplus_r W_r^{p_r}$ as H -representations, and $\text{Ind}_H^G W_j \cong \bigoplus_s V_s^{q_s}$ as G -representations, then using the orthonormality of the character (together with the fact that direct sum representation has character, being the sum of the direct summands' characters), one can deduce that each $p_r = \langle \chi_{\text{Res}_H^G V_i}, \chi_{W_r} \rangle_H$, and $q_s = \langle \chi_{\text{Ind}_H^G W_j}, \chi_{V_s} \rangle_G$ (with $p_r, q_s \in \mathbb{N}$). Hence, one has the following, based on Schur's Lemma:

$$\begin{aligned}
\mathrm{Hom}_G(V_i, \mathrm{Ind}_H^G W_j) &\cong \mathrm{Hom}_G\left(V_i, \bigoplus_s V_s^{q_s}\right) \cong \bigoplus_s \mathrm{Hom}_G(V_i, V_s)^{q_s} \cong \mathrm{Hom}_G(V_i, V_i)^{q_i} \cong \mathbb{C}^{q_i} \\
\mathrm{Hom}_H(\mathrm{Res}_H^G V_i, W_j) &\cong \mathrm{Hom}_H\left(\bigoplus_r W_r^{p_r}, W_j\right) \cong \bigoplus_r \mathrm{Hom}_H(W_r, W_j)^{p_r} \cong \mathrm{Hom}_H(W_j, W_j)^{p_j} \cong \mathbb{C}^{p_j}
\end{aligned} \tag{2.50}$$

Finally, using the inner product formula (relating Res and Ind), one has the following:

$$q_i = \langle \chi_{\mathrm{Ind}_H^G W_j}, \chi_{V_i} \rangle_G = \langle \chi_{V_i}, \chi_{\mathrm{Ind}_H^G W_j} \rangle_G = \langle \chi_{\mathrm{Res}_H^G V_i}, \chi_{W_j} \rangle_H = p_j \tag{2.51}$$

Hence, $\mathrm{Hom}_G(V_i, \mathrm{Ind}_H^G W_j) \cong \mathbb{C}^{q_i} = \mathbb{C}^{p_j} \cong \mathrm{Hom}_H(\mathrm{Res}_H^G V_i, W_j)$, this finishes the proof.

3 Bonus:

Problem 12

Continue to let $W \in \text{Rep}_k(H)$ and let $V \in \text{Rep}_k(G)$. Let $f : W \rightarrow V$ be a map of H -representations. Let us define a map of vector spaces $g_f^{(i)} : W^{(i)} \rightarrow V$ as follow: we define

$$g_f^{(i)}(\varphi_w^{(i)}) := x_i \cdot f(w) \in V \quad (3.1)$$

where the action by x_i is given using the G -representation structure on V . This way (by Part (3)) we obtain a map $g_f : \text{Ind}_H^G W \rightarrow V$ as vector spaces. Show that g_f is in fact a map between G -representations. Use this to show that Ind_H^G is also a left adjoint of Res_H^G when G is finite.

Solution:

Problem 13

Show that $V = \text{Ind}_H^G W$ is irreducible if and only if the following holds:

- W is irreducible
- For any $s \in G \setminus H$, we have

$$\langle \rho_s, \text{Res}_H^G \rho \rangle_{H_s} = 0 \quad (3.2)$$

where $H_s := sHs^{-1} \cap H$ and ρ_s denotes the representation of H_s on W given by $\rho_s(x) = \rho_W(sxs^{-1})$.